

NIHARIKA BHASIN

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Education

New York University

Masters in Computer Science

Sep 2024 – May 2026

Brooklyn, NY

SRM Institute of Science and Technology

Bachelors of Technology in Computer Science Engineering

Sep 2020 – May 2024

Delhi, India

Technical Skills

Languages & Platforms: Python, C/C++, Java, JavaScript, SQL, Linux, Embedded Platforms

Systems & Architecture: Distributed Systems, Microservices, Event-Driven Architecture, Agile Software Development

Cloud & Infrastructure: AWS (IAM, Lambda, DynamoDB, EC2, S3), Docker, Kubernetes, Terraform, Jenkins, Git

Data, Backend & Streaming: FastAPI, Django, Node.js, REST APIs, WebSockets, PostgreSQL, PostGIS, Redis, Kafka

Geospatial & Algorithms: Computational Geometry, Spatial Indexing (R-Tree), Routing Systems, GIS Data, Mapbox

Experience

Software Engineering GA, NYU ECE

June 2025 – May 2026

New York University

Brooklyn, NY

- Engineered a centralized **Lab Operations Portal** using **FastAPI** and **React**, replacing fragmented manual logs for **150+ researchers** and automating equipment scheduling with real-time inventory tracking.
- Architected high-throughput **RESTful APIs** and debugged search bottlenecks via **PostgreSQL** indexing and query optimization, reducing **P99 latency by 45%** during peak academic usage.
- Designed and deployed automated **ETL pipelines** using **Python** and Pandas to synchronize database records, eliminating **100% of redundant workflows** and cutting administrative processing time by **35%**.
- Hardened application security by implementing granular **RBAC** and custom validation middleware, maintaining **99.99% data integrity**; authored technical documentation to support software design requirements.

Software Engineering Intern

June 2025 – August 2025

Toricient Labs

New York City, USA

- Architected a massively scalable **serverless backend on AWS (Lambda, API Gateway, DynamoDB)**, engineering robust software functionality for high-throughput workloads exceeding **1M+ monthly API invocations**.
- Spearheaded a **zero-downtime infrastructure migration** to AWS Cloud, utilizing **Terraform (IaC)** to slash **operational expenditure by 30%** and eliminate vendor lock-in.
- Developed **50+ production-grade microservices in Node.js and Python**, participating in **code reviews** to ensure quality standards and integrating **OAuth2** authentication for enterprise clients.
- Debugged distributed caching performance issues by implementing **Redis (DAX)**, reducing **P99 API latency by 25%** under peak stress tests.
- Maintained **99.9% system uptime** via event-driven workflows and full-stack observability using **AWS CloudWatch**; collaborated with QA to verify deployments met technical specifications, and authored documentation covering **API contracts**, infrastructure design, and deployment runbooks for cross-team knowledge sharing.

Projects

SafeZone — Geospatial Navigation & Routing Engine (Django REST, PostGIS, Mapbox)

- Architected a containerized full-stack geospatial navigation system using **PostgreSQL/PostGIS** for high-performance spatial indexing; debugged routing calculation overhead, achieving a **40% reduction** via Mapbox engine optimization.
- Secured application access using **AWS IAM** via **AWS Cognito RBAC**; collaborated with QA to implement **Playwright E2E testing** protocols, verifying applications meet user needs and technical specifications.

CodeSync — Real-Time Distributed System & Stream Processing (WebSockets, AWS, Docker)

- Built a low-latency real-time code collaboration engine using **WebSockets** and **AWS SQS**, achieving **sub-50ms sync delay** across concurrent users to optimize cross-functional teamwork.
- Deployed **multi-tenant Docker sandboxes** on isolated **EC2** nodes via automated **CI/CD pipelines**; authored technical documentation covering system design, API specifications, and deployment procedures.

Ticket Finder — High-Concurrency Distributed Platform (FastAPI, Redis, PostgreSQL)

- Designed a **high-concurrency distributed system** handling **10k+ concurrent requests** within an Agile software development workflow, ensuring application reliability under heavy stress loads.
- Implemented **continuous testing protocols** to identify bugs and race conditions; debugged software layers via **dual-layer concurrency control**, enhancing performance and overall user experience.